Simulation of Solar Wind Charged Particle Energy Deposited and Particle Identification by ΔE-E Discrimination in the SNAPPY CubeSat Detector

Daniel Reichart, NASA Jump Start Student
Wichita State University, Department of Aerospace Engineering and Physics

Introduction

What is vSOL?

- The focus of the neutrino Solar Orbiting Laboratory (vSOL) is the detection and study of neutrinos, primarily from the sun.
- vSOL seeks to utilize the increased neutrino flux at distances close to the Sun in order to:
  - Better detect neutrinos
  - Understand the Sun's fusion core
  - Come to a better understanding of neutrino osciallations
- Neutrinos are elementary particles that are extremely light and electrically neutral. Due to these properties, neutrinos are notoriously difficult to detect and study.
- vSOL's overall goal is to present a new and innovative way of studying the sun's fusion core and the particle physics of neutrinos.

Secondary Study

- The aim of my research is finding out how well particles, other than neutrinos, can be differentiated with the CubeSat Detector.
- Further efforts are being made in determining not only the particle type, but also the particle energy.

Detectors

The GAGG

- GAGG stands for gadolinium aluminum gallium garnet, which is the material that the inner detector is made from.
- This material was chosen because it has a very recognizable interaction with neutrinos that is characterized by a double pulse.

The Veto

- The Veto gets its name because of its purpose in the CubeSat Detector. If a signal is captured by the Veto at the same time as the GAGG, then the interaction is not considered a neutrino interaction because neutrinos have a very low chance of interacting with the Veto's material.

ΔE-E Method

ΔE Detector

- The Veto will act as the ΔE detector in the set up for this method. The ΔE detector is usually the first detector in a string of detectors, the purpose of which is to collect energy deposits from incoming particles.

E Detector

- The GAGG will act as the E detector for this method. The E detector is usually the final detector in the string of detectors, but in this case is enclosed within the Veto (ΔE detector). This will create deviations from usual ΔE-E plots.

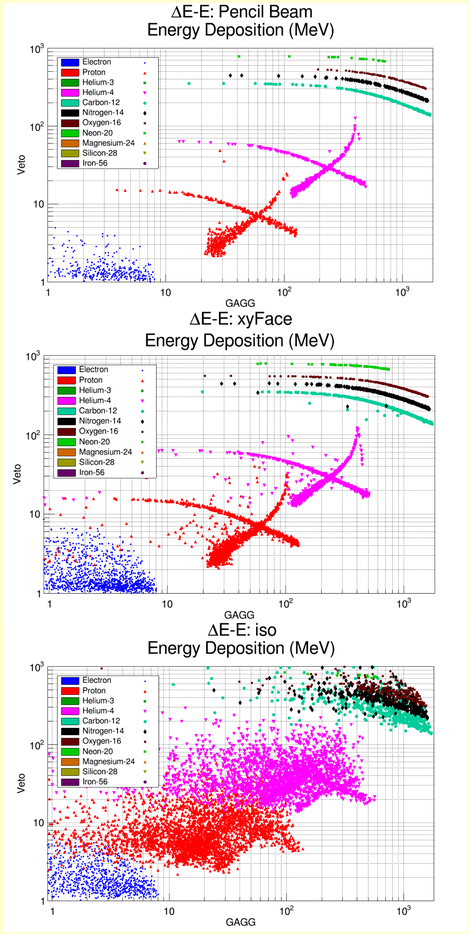
ΔE-E Plot

- The ΔE detector's energy deposits are plotted on the vertical axis, while the E detector's is plotted on the horizontal axis. The point (E, ΔE) is the energy deposit from a single particle event.
- This method allows for distinct particle regions due to their distinct particle mass, that allows us to identify which particles are detected based on their location on the plot.

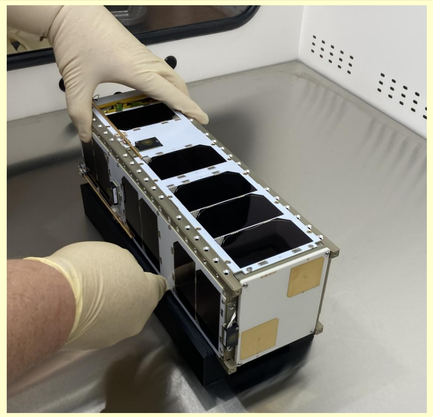
Relevance to Kansas

Solar wind particles can be hazardous to both energy generation and transmission systems on the ground as well as to aviation flight. Identifying particles in solar wind can further our understanding of these hazards and help determine solutions.

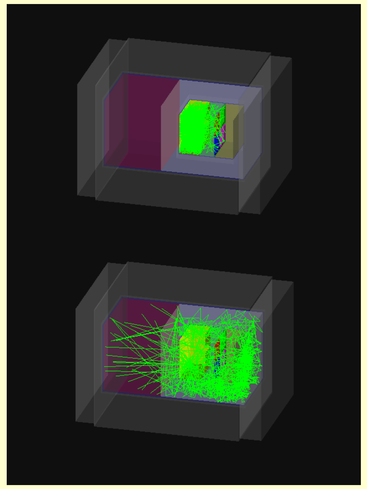
Simulation Results



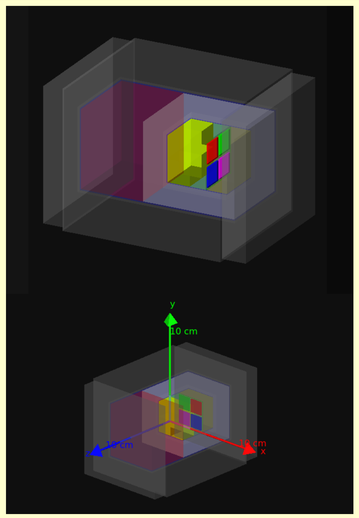
CubeSat Nanosatellite



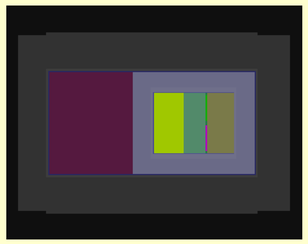
Energy Deposit in GEANT4



Detector Geometry



- Dark Gray Volume – Tungsten-Epoxy Shielding
  - Functions as a lightweight radiation shield, that burns up during reentry.
- Maroon Volume – The Veto's Photomultiplier Tube
  - Collects the scintillation light produced in the Veto.
- Light Blue Volume – The Veto
- Yellow Volume – The GAGG
- Dark Green Volume – Quartz Crystal Window
  - Added for optical properties
- Light Green, Purple, Dark Blue, and Red Volumes – The GAGG's Photomultiplier Tubes
  - Collects the scintillation light produced in the GAGG
- Brown Volume – Electronics for the GAGG
- The red arrow pointed to the right is the positive x axis.
- The green arrow pointed up is the positive y axis.
- The blue arrow pointed to the left is the positive z axis.



References

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- McGuire, R., & Von Rosenvinge, T. (1984). The energy spectra of solar energetic particles. Advances in Space Research, 4(2–3), 117–125.
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Simulation Scenarios

- **Pencil Beam**

The purpose of this scenario is to examine how the CubeSat Detector would perform if experimented on with a particle accelerator. The particles are fired in the same place for every event, where they will penetrate the Veto and GAGG from the shortest possible distance. The pencil beam is effectively the most ideal scenario.

- **Face Scenarios**

The purpose of the face simulations is to acquire a direct comparison between general ΔE-E telescopes and the CubeSat Detector. The flight path of the particles are similar to the pencil beam; however, their positions vary across an entire face of the detector.

- **Impingement of the x and y Faces**
  - Because of the symmetry of the CubeSat Detector, the ± x and ± y faces are identical. xyFace directs the particles at the + x face, but these results apply to all four faces.
- **Impingement of the + z Face**
  - z+Face fires particles at the + z face of the detector.
- **Impingement of the – z Face**
  - z-Face fires particles at the – z face of the detector.

Space Flight Scenarios

These scenarios are similar to the conditions the CubeSat will be under while in space. The particles are fired in a random flight path (specifically isotropic), mimicking that of the particles on solar wind.

- **Hemispherical Impingement of the + z Face**
  - hemisphereIsoZ+Face fires the particles at the + z face of the detector from the surface of an imaginary half sphere that partially surrounds the + z face of the detector.
- **Hemispherical Impingement of the - z Face**
  - hemisphereIsoZ-Face fires the particles at the - z face of the detector from the surface of another imaginary half sphere that partially surrounds the - z face of the detector.
- **Spherical Impingement of the CubeSat Detector**
  - iso combines the strategies of the previous isotropic scenarios, firing particles from the surface of an imaginary sphere that surrounds the CubeSat Detector. Only a reference scenario.

Particles/Ions and Energy Ranges

e^- ---------- 1 MeV to 10.15 MeV

${}^{1}_{1}H^{+}$ ---------- 1 MeV to 2.55 GeV

${}^{3}_{2}He^{+2}$ ---------- 3 MeV to 19.50 MeV

${}^{4}_{2}He^{+2}$ ---------- 4 MeV to 3.16 GeV

${}^{12}_{6}C^{+6}$ ---------- 12 MeV to 1.95 GeV

${}^{14}_{7}N^{+7}$ ---------- 14 MeV to 1.99 GeV

${}^{16}_{8}O^{+8}$ ---------- 16 MeV to 2.08 GeV

${}^{20}_{10}Ne^{+10}$ ---------- 20 MeV to 1.19 GeV

${}^{24}_{12}Mg^{+12}$ ---------- 24 MeV to 1.95 GeV

${}^{28}_{14}Si^{+14}$ ---------- 28 MeV to 1.94 GeV

${}^{56}_{26}Fe^{+26}$ ---------- 56 MeV to 1.58 GeV

Conclusion

The heavier Ions become bundled together and are not fully distinguishable. However, electrons, protons, and alpha particles all have regions of separation that are large enough to tell them apart. This suggests that particle identification via ΔE-E is a viable option for electrons, protons, and alpha particles in the CubeSat Detector.